

The multidimensional driving style inventory—scale construct and validation

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Abstract

Two studies were conducted in order to develop a multidimensional instrument of driving style. In Study 1, we developed a self-report scale assessing four broad domains of driving style—the multidimensional driving style inventory (MDSI). A factor analysis revealed eight main factors, each one representing a specific driving style—dissociative, anxious, risky, angry, high-velocity, distress reduction, patient, and careful. In addition, significant associations were found between the eight factors, on the one hand, and gender, age, driving history, and personality measures of self-esteem, need for control, impulsive sensation seeking, and extraversion, on the other. In Study 2, further associations were found between the eight driving style factors and measures of trait anxiety and neuroticism. The discussion focused on the validity and utility of a multidimensional conceptualization of driving style.

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1. Introduction

In the last several years, there has been a growing concern about the harsh consequences of driving and an increased level of interest in the traffic safety problem of car accidents (Harré, 2000; West et al., 1993). This line of research has mainly focused on human factors that are involved in car accidents, such as sociodemographic and general personality factors as well as driving-specific skills, attitudes, and behaviors (e.g. Beirness, 1993; Garrity and Demick, 2001; Jonah, 1997; West et al., 1993). The current study follows this line of research and mainly focuses on the conceptualization of a person's habitual driving style as a driving-specific factor that can directly explain involvement in car accidents and mediate the effects of more general sociodemographic and personality factors.

A review of the literature indicates that previous research has mostly dealt with the association between various sociodemographic factors (e.g. age, gender, experience) or general personality traits (e.g. sensation seeking, type A/B personality, locus of control) and involvement in car accidents (e.g. Furnham and Saipe, 1993). In this context, gender and age have consistently been found to be related to driver's

accident risk and traffic violations (e.g. Lawton et al., 1997; Maycock et al., 1991; Westerman and Haigney, 2000). Almost every measure of involvement in fatal crashes recorded in the USA during the 1980s showed rates for men approximately double those for women (Evans, 1991), as well as increased crash involvement and a higher rate of risk taking while driving among younger drivers (e.g. Glendon et al., 1996; Matthews and Moran, 1986; Maycock et al., 1991).

Personality traits have been also shown to be related to risky driving and crash involvement. In this context, traits of sensation seeking, impulsiveness, and thrill and adventure seeking seem to be the strongest predictors of reckless driving and involvement in car accidents. Specifically, these traits have been consistently associated with engagement in risky driving practices, such as speeding or impaired driving, and involvement in traffic violations and accidents (e.g. Arnett et al., 1997; Beirness and Simpson, 1988, 1990; Donovan et al., 1990; Jonah, 1997; Trimpop and Kirkcaldy, 1997; Zuckerman and Neeb, 1980). Accordingly, some studies have reported that the trait of desire for control is also related to reckless driving and car accidents (e.g. Horswill and McKenna, 1999; Trimpop and Kirkcaldy, 1997). With regard to traits of extraversion and neuroticism (Eysenck and Eysenck, 1975), the findings are less conclusive. On the one hand, some studies have found significant associations between these traits and both number of crashes and violations

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(e.g. Fine, 1963; Renner and Anderle, 1999; Shaw and Sichel, 1971). On the other hand, these findings were significant only for men and additional studies have failed to find such a relationship even among men (e.g. Matthews et al., 1991; Wilson and Greensmith, 1983).

To date, most of the researchers agree that the above reviewed findings are highly important for understanding involvement in car accidents, but they do not provide information about the specific driving-related factors that directly underlie reckless driving. In this context, Elander et al. (1993) have argued that accident liability is related to driving skill and to driving style. By “skill” they referred to the abilities of drivers to maintain control of the vehicle and respond adaptively to complex traffic situations. In other words, they refer to the driver’s performance. Driving skill is expected to improve with practice or training. By “style” they referred to the ways drivers choose to drive or habitually drive. This includes choice of driving speed, headway, and habitual level of general attentiveness and assertiveness. Driving style is expected to be influenced by attitudes and beliefs regarding driving as well as more general needs and values. It is this aspect of driving that stands in the focus of the present investigation.

Despite the agreement about the importance of driving style, there is no agreement about its conceptualization and measurement. In fact, several self-report measures of driver behavior and cognition tapping very different aspects of driving have been constructed in the last years (Westerman and Haigney, 2000)—Driving Behavior Inventory (DBI, Gulian et al., 1988; Gulian et al., 1989), Driving Style Questionnaire (DSQ, French et al., 1993), The Attitudes to Driving Violations (ADVS, West and Hall, 1997), Driver Behavior Questionnaire (DBQ, Reason et al., 1990), Drivers Behavior Questionnaire (Furnham and Saiepe, 1993), and Driving Vengeance Questionnaire (DVQ, Wiesenthal et al., 2000).

The DSQ (French et al., 1993), for example, examines behaviors that had been shown to be related to accident involvement or risky driving behavior, such as, speed, headway (distance to the car in front), seat belt use, gap acceptance (size of gap in the flow of traffic before attempting to pull out), and traffic light violations, as well as cognitions and attitudes that are supposed to be directly related to driving decision-making, such as feelings of control, route planning, and risk taking on the road. Another assessment procedure, the DBI (Gulian et al., 1988, 1989), focuses on driving stress and taps dimensions of driving aggression, driving alertness, dislike of driving, general driver stress, irritation when overtaken, and frustration in overtaking. Yet, the DBQ (Reason et al., 1990) examines errors made while driving, deliberate violations of normal safe driving practice, and harmless mistakes that result from inattention (lapses).

We believe that this diversity of conceptualizations and measurement scales tapping driving style reflects the highly complex and multidimensional nature of the phenomenon. However, we also think that the status of theory and research in driving style enables the conceptual and empirical inte-

gration of the various definitions and scales into a single, multidimensional conceptualization of driving style. On this basis, we reviewed the diverse scales of driving styles and conceptually analyzed the underlying factor structures of these scales. Even though most researchers were interested in driving behaviors which are related to car accidents, we broadened our scope to various behaviors and habits which are related to driving in general in order to reveal the whole range of driving styles that can predict involvement in car accidents.

Following a review of the existing scales of driving styles, we hypothesize that most of the driving-specific factors identified in these scales can be integrated into four broad facets: (a) reckless and careless driving style, (b) anxious driving style, (c) angry and hostile driving style, (d) patient and careful driving style. The *reckless and careless driving style* refers to deliberate violations of safe driving norms, and the seeking of sensations and thrill while driving (e.g. French et al., 1993; Reason et al., 1990). It characterizes persons who drive at high speeds, race in cars, pass other cars in no-passing zones, and drive while intoxicated, probably endangering themselves and others. The *anxious driving style* has commonly been examined in studies on driver stress (e.g. Gulian et al., 1989) and reflects feelings of alertness and tension as well as ineffective engagement in relaxing activities during driving. The *angry and hostile driving style* refers to expressions of irritation, rage, and hostile attitudes and acts while driving, and reflects a tendency to act aggressively on the road, curse, blow horn, or “flash” to other drivers (e.g. Arnett et al., 1997; Donovan et al., 1988). The *patient and careful driving style* reflects a well-adjusted driving style that has received less attention in previous studies (e.g. French et al., 1993; Harré, 2000). This style refers to planning ahead; attention, patience, politeness, and calmness while driving; and keeping the traffic rules.

After conceptualizing the above four domains of driving styles, our next steps were to build a self-report scale for assessing these domains, to examine whether the factor structure of this scale validated the hypothesized four domains, and to explore the associations between these domains, driving behaviors, and sociodemographic and personal factors. This research program can provide important information on the usefulness of a multidimensional scale for assessing driving styles and clarify the associations of these styles with a host of other variables.

2. Study 1

The first study was intended to construct a self-report instrument in order to assess the four domains of driving style and their relevance for examining variations in history of driving in general and reckless driving in particular. We drew on existing theoretical and empirical literature to identify four domains of driving style. Next, we constructed a measure to assess driving styles in these domains

by adapting items from several existing measures, such as the DSQ (French et al., 1993), DBQ (Reason et al., 1990), DBI (Gulian et al., 1988, 1989), and by writing additional original items. Then, we examined the associations between these styles and measures of reckless driving, sociodemographic factors (gender, age, level of education), and personality traits (self-esteem, need for control, impulsive sensation seeking, and extraversion). Since previous studies have generally failed to take into account annual mileage, which has been found related to accident rates and to propensity to drive fast (e.g. French et al., 1993; Quimby et al., 1986), we controlled for variations in annual mileage while examining the above associations.

2.1. Method

2.1.1. Participants

Three hundred and twenty eight participants from various geographical areas in Israel who owned a driving license and drove on a regular basis volunteered to participate in the study. These participants were sampled via the “snowball” technique: the questionnaires were given to an initial sample of university and college students, who asked friends, acquaintances, and family members to complete the questionnaire. The sample consisted of 220 women and 108 men, ranging in age from 19 to 70 (mean = 31.78, S.D. = 13.31). Sixty-two percent of them ($N = 189$) were university students, 12% completed elementary school, and 26% completed high school education.

2.1.2. Procedure and measures

Participants were asked to complete a packet of questionnaires. The questionnaires were presented in a random order across participants. The packet included scales tapping driving style, self-esteem, desire for control, impulsive sensation seeking, extraversion, and driving behaviors.

Driving style was assessed by the *multidimensional driving style inventory (MDSI)*, which has been especially constructed for this study in order to tap the four hypothesized domains of driving styles. Participants were asked to read each item and to rate the extent to which it fits their feelings, thoughts, and behavior during driving on a 6-point scale, ranging from “not at all” (1) to “very much” (6).¹ Originally, 20 items were written to assess each of the four domains. This 80-item version was administered to a pilot sample of 500 participants (354 women and 146 men, ranging in age from 19 to 42, median = 28), most of them university students. Following item and exploratory factor analyses, we retain 44 items that have an adequate normal distribution and good psychometric features. These 44 items became the final version of the MDSI and all the statistical analyses were conducted on this version of the scale.

Global self-esteem was assessed by Rosenberg’s (1979) 10-item scale. Participants rated their agreement with each item on a 4-point scale, ranging from 1 (*strongly disagree*) to 4 (*strongly agree*). In the current sample, the Cronbach’s α for the 10 items was high (0.86). Then, we averaged the 10 items, with higher scores indicating more positive self-esteem. Desire for control was assessed by Burger and Cooper’s (1979) 20-item scale, which taps need for control in daily activities. Participants are required to respond on a 7-point Likert type scale, ranging from 1 (*never*) to 7 (*always*). The Cronbach’s α for the 20 items was acceptable (0.77). Thus, we averaged all items into a single score, with higher scores representing higher desire for control.

Impulsive sensation seeking was assessed by Zuckerman et al. (1993) 19-item scale, which taps needs for stimulation and sensation as well as impulsiveness and risk taking in decision-making. The Cronbach’s α for the 19 items was acceptable (0.80), thus we averaged all items into a single score, with higher scores representing higher impulsive sensation seeking. Extraversion was assessed by the Extraversion subscale of the Eysenck Personality Inventory (Eysenck and Eysenck, 1967), which was composed of 23 items that could be answered *yes* or *no*. The Cronbach’s α for the 23 items was acceptable (0.79). Thus, we averaged all items into a single score, with higher scores representing higher extraversion.

At the end of these questionnaires, participants were asked to provide sociodemographic information as well as information about exposure, by reporting on the average amount of kilometers driven per day during the week and during weekends; involvement in car accidents (the lifetime number of involvement in car accidents), and the lifetime frequency of 13 driving offenses (e.g. speeding, crossing in red light).

2.2. Results and discussion

2.2.1. MDSI factors

To determine whether the 44 MDSI items fell into distinguishable domains, a factor analysis with Varimax rotation was conducted on these 44 items. The factor analysis revealed eight main factors (eigenvalue > 1), which explained 56% of the variance of the 44 items. Table 1 presents loadings of the items in each of the factors. Factor 1 explained 21% of the variance (Cronbach’s $\alpha = 0.82$) and consisted of 8 items that load high (greater than 0.40) on the factor. All these items tap a person’s tendency to be easily distracted during driving, to commit driving errors due to this distraction, and to display cognitive gaps and dissociations during driving. On this basis, we labeled this factor as “dissociative driving style”. Factor 2 explained 10% of the variance (Cronbach’s $\alpha = 0.82$) and consisted of 7 items that load high on the factor. All these items tap a person’s tendency to feel distress during driving, to display signs of anxiety due to the driving situation, and to express doubts and lack of confidence about his or her driving skills. On this basis, we labeled this factor as “anxious driving style”.

¹ The full questionnaire can be found on the web at: <http://members.tripod.com/drive10/>.

Table 1

Factor model coefficients of the multidimensional driving style inventory

Factors and items	Loading
Factor 1: dissociative driving style	
[30] misjudge the speed of an oncoming vehicle when passing	0.76
[34] intend to switch on the windscreen wipers, but switch on the lights instead	0.70
[27] forget that my lights are on full beam until flashed by another motorist	0.69
[39] nearly hit something due to misjudging my gap in a parking lot	0.68
[36] plan my route badly, so that I hit traffic that I could have avoided	0.56
[35] attempt to drive away from traffic lights in third gear (or on the neutral mode in automatic cars)	0.48
[15] lost in thoughts or distracted, I fail to notice someone at the pedestrian crossings	0.48
[11] I daydream to pass the time while driving	0.47
Factor 2: anxious driving style	
[31] feel nervous while driving	0.75
[33] feel distressed while driving	0.75
[10] driving makes me feel frustrated	0.68
[25] it worries me when driving in bad weather	0.52
[7] on a clear freeway, I usually drive at or a little below the speed limit	0.52
[4] feel I have control over driving [–]	0.49
[40] feel comfortable while driving [–]	0.48
Factor 3: risky driving style	
[44] enjoy the excitement of dangerous driving	0.83
[6] enjoy the sensation of driving on the limit	0.82
[22] like to take risks while driving	0.80
[24] like the thrill of flirting with death or disaster	0.66
[20] fix my hair/ makeup while driving	0.45
Factor 4: angry driving style	
[12] swear at other drivers	0.72
[3] blow my horn or “flash” the car in front as a way of expressing frustrations	0.72
[28] when someone does something on the road that annoys me, I flash them with the high beam	0.73
[43] honk my horn at others	0.67
[19] when someone tries to skirt in front of me on the road, I drive in an assertive way in order to prevent it	0.48
Factor 5: high-velocity driving style	
[16] in a traffic jam, I think about ways to get through the traffic faster	0.72
[9] when in a traffic jam and the lane next to me starts to move, I try to move into that lane as soon as possible	0.71
[17] when a traffic light turns green and the car in front of me doesn’t get going immediately, I try to urge the driver to move on	0.59
[2] purposely tailgate other drivers	0.58
[32] get impatient during rush hours	0.46
[5] drive through traffic lights that have just turned red	0.40
Factor 6: distress-reduction driving style	
[37] use muscle relaxation techniques while driving	0.73
[8] while driving, I try to relax myself	0.71
[1] do relaxing activities while driving	0.63
[26] mediate while driving	0.56
Factor 7: patient driving style	
[18] at an intersection where I have to give right-of-way to oncoming traffic, I wait patiently for cross-traffic to pass	0.68
[23] base my behavior on the motto “better safe than sorry”	0.52
[13] when a traffic light turns green and the car in front of me doesn’t get going, I just wait for a while until it moves	0.49
[38] plan long journeys in advance	0.49
Factor 8: careful driving style	
[42] tend to drive cautiously	0.56
[14] drive cautiously	0.55
[41] always ready to react to unexpected maneuvers by other drivers	0.51
[21] distracted or preoccupied, and suddenly realize the vehicle ahead has slowed down, and have to slam on the breaks to avoid a collision [–]	0.51
[29] get a thrill out of breaking the law [–]	0.51

Numbers in brackets represent the order of the items in the scale.

[–] reversed item.

The detailed loadings of the 44 items on the 8 factors are available upon request from the authors.

Factor 3 explained 6% of the variance (Cronbach's $\alpha = 0.83$) and consisted of 5 items that load high on the factor. All these items tap a person's seeking for stimulation, sensation, and risk during driving and his or her tendency to take risky driving decisions and to engage in risky driving. On this basis, we labeled this factor as "risky driving style". Factor 4 explained 5% of the variance (Cronbach's $\alpha = 0.80$) and consisted of 5 items that load high on the factor. All the items tap a person's tendency to be hostile towards other drivers as well as behave aggressively and feel intense anger while driving. On this basis, we labeled this factor as "angry driving style". Factor 5 explained 4% of the variance (Cronbach's $\alpha = 0.76$) and consisted of 6 items that load high on the factor. All the items tap a person's tendency to drive fast, to display signs of time pressure while driving, and to be oriented towards high velocity driving. Therefore, we labeled this factor as "high-velocity driving style".

Factor 6 explained 4% of the variance (Cronbach's $\alpha = 0.75$) and consisted of 4 items that load high on the factor. These items tap a person's tendency to engage in relaxing activities during driving aimed at reducing distress while driving. On this basis, we labeled this factor as "distress-reduction driving style". Factor 7 explained 3% of the variance (Cronbach's $\alpha = 0.74$) and consisted of 4 items that load high on the factor. All the items tap a person's tendency to be polite towards other drivers, to feel no time pressure during driving, and to display patience while driving. On this basis, we labeled this factor as "patient driving style". Factor 8 explained 3% of the variance (Cronbach's $\alpha = 0.76$) and consisted of 5 items that load high on the factor. All the items tap a person's tendency to be careful during driving, to effectively plan his or her driving trajectory, and to adopt a problem-solving attitude towards driving-related problems and obstacles. On this basis, we labeled this factor as "careful driving style".

As can be seen, the factor analysis revealed eight coherent and meaningful driving styles. Scores for each of the eight factors were computed by averaging items loading high on each factor. Pearson correlations between the eight factors revealed an interesting pattern of associations. First, significant positive associations were found between risky, high-velocity, angry, and dissociative driving styles, $r(s)$ ranging from 0.34 to 0.50, all $P(s) < 0.01$, implying the existence of an underlying maladaptive way of driving that may be theoretically associated with emotional maladjustment as well as with high likelihood of car accidents and driving offenses. Second, the above four maladaptive driving styles were inversely and significantly associated with the careful and patient styles, that reflect more adequate, controlled, and socially adjusted ways of driving, $r(s)$ ranging from -0.20 to -0.49 , all $P(s) < 0.01$. Third, significant positive associations were found between the careful and patient styles, $r(309) = 0.21$, $P < 0.01$, as well as between the anxious and distress-reduction factors, $r(309) = 0.25$, $P < 0.01$. Fourth, the anxious and dissociative styles

were also significantly associated, $r(309) = 0.47$, $P < 0.01$. Other correlations were not statistically significant.

Overall, the MDSI presents a comprehensive, multidimensional picture of the various orientations people may adopt while driving. In this way, the MDSI compliments existing self-report scales. Whereas these scales focus on only one or two of the MDSI factors (e.g. driving stress, driving aggression, risky driving), the MDSI could delineate a person's profile across eight differentiated, and even antagonistic, driving orientations.

2.2.2. Driving styles and sociodemographic factors

In the next step, we examined the association between the eight driving style scores and three basic sociodemographic characteristics (sex, age, education level). Gender differences in driving style were examined by multivariate and univariate analyses of variance (ANOVA). The multivariate ANOVA revealed a significant gender difference, $F(8, 319) = 5.39$, $P < 0.01$. Univariate ANOVAs indicated that gender differences were significant in dissociative driving style, $F(1, 326) = 14.74$, $P < 0.01$, anxious driving style, $F(1, 326) = 10.77$, $P < 0.01$, and careful driving style, $F(1, 326) = 24.13$, $P < 0.01$. An examination of group means (see Table 2) revealed that women scored higher in dissociative and anxious driving styles than men. Men scored higher than women in careful driving style. The

Table 2
Means and S.D. of the multidimensional driving style inventory factors according to gender

MDSI factors	Men ($n = 108$)	Women ($n = 220$)
Dissociative		
Mean	1.80	2.13
S.D.	0.57	0.74
Anxious		
Mean	2.02	2.35
S.D.	0.72	0.83
Risky		
Mean	1.47	1.45
S.D.	0.71	0.73
Angry		
Mean	2.45	2.32
S.D.	1.04	0.93
High-velocity		
Mean	3.02	2.92
S.D.	0.88	0.87
Distress reduction		
Mean	2.31	2.48
S.D.	0.94	0.82
Patient		
Mean	4.74	4.72
S.D.	0.97	0.68
Careful		
Mean	4.60	4.19
S.D.	0.65	0.68

same gender differences were obtained after controlling for the amount of weekly driving.

Pearson correlations between age and the eight driving style scores revealed the following significant associations: age was positively associated with careful and patient driving styles, $r(326) = 0.17$, $P < 0.01$; $r(326) = 0.40$, $P < 0.01$, and inversely associated with dissociative, angry, anxious, risky, and high-velocity driving, $r(326) = -0.39$, $P < 0.01$; $r(326) = -0.20$, $P < 0.01$, $r(326) = -0.22$, $P < 0.01$; $r(326) = -0.26$, $P < 0.01$; $r(326) = -0.19$, $P < 0.01$. That is the older the participant, the higher his or her tendency to adopt a careful and patient driving style and the lower his or her tendency to adopt dissociative, angry, anxious, risky, or high-velocity driving styles. These correlations remained the same after controlling for the amount of weekly driving.

Partial correlations between education level and the eight driving style scores (controlling for age) revealed significant associations between education level and the endorsement of anxious and distress-reduction driving styles, $r(326) = 0.18$, $P < 0.01$; $r(326) = 0.18$, $P < 0.01$. That is the higher the education level of a participant, the higher his or her tendency to feel anxiety during driving and to adopt a distress-reduction style.

Overall, the findings strengthen the confidence in the construct validity of the DSI factors. Our findings were in accordance with the literature, which reveals that women tend to exhibit more driving stress than men and that maladaptive driving seems to diminish with age. These two tendencies were clearly identified by the DSI factors. First, women's driving stress was manifested in their relatively high scores in anxious and dissociative driving styles. Second, the tendency of older people to adopt more adaptive ways of driving was manifested in their relatively high scores in careful and patient driving styles as well as in their relatively low scores in angry, anxious, dissociative, risky, and high-velocity driving styles.

2.2.3. Driving styles and personality traits

A canonical correlation between the set of the eight driving style scores and the set of the four assessed personality traits revealed a significant association, $F(32, 717) = 4.64$, $P < 0.01$ and explained 38% of the variance. Pearson correlations (see Table 3) revealed the following significant associations: first, self-esteem was significantly and positively associated with careful and patient driving styles, and inversely associated with dissociative and risky driving styles. That is the more positive a participant's self-esteem, the higher his or her tendency to adopt a careful and patient driving style and the lower his or her tendency to adopt dissociative or risky driving styles. This pattern of findings strengthens our confidence in the validity of the MDSI as measuring adaptive and maladaptive driving styles. Self-esteem is viewed as one of the basic signs of psychological adjustment and then should be positively associated with well-adjusted styles of driving and inversely associated

Table 3

Pearson correlations between driving style inventory factors and personality traits

MDSI factors	Self-esteem	Need for control	Sensation seeking	Extraversion
Dissociative	−0.38**	−0.04	0.10	−0.23**
Anxious	−0.05	−0.08	−0.11	−0.22*
Risky	−0.19*	0.09	0.40**	−0.02
Angry	−0.10	0.22*	0.13	0.14
High-velocity	−0.11	0.13	0.18*	0.01
Distress reduction	0.04	0.01	0.01	0.06
Patient	0.23**	−0.04	−0.09	−0.16
Careful	0.27**	0.17*	−0.31**	0.02

* $P < 0.01$.

** $P < 0.001$.

with maladjusted ways of driving. As can be seen, the current findings provide strong support for this hypothesis. Whereas self-esteem was positively associated with careful and patient driving styles, the two adaptive driving styles, it was inversely associated with dissociative and risky driving, which are considered maladaptive driving styles.

Second, need for control was significantly and positively associated with angry and careful driving styles. That is the higher the need for control, the higher the tendency to adopt an angry or careful driving style. This pattern of findings reflects that one of the psychological sources of angry driving style is a strong need for control and that the frustration of such a need during driving could result in anger, aggression, and hostility towards other drivers. Interestingly, need for control seems also to underlie careful driving style. This is an expected finding because careful driving has a planning, problem-solving facet (see items in Table 1), implying that the driver feels that driving is under his or her control. On this basis, we can conclude that desire for control may have both positive and negative driving consequences. Whereas it could lead to a more careful driving, its frustration could lead to angry driving.

Third, sensation seeking was significantly and positively associated with risky and high-velocity driving styles, and inversely associated with patient driving. As expected, a person's global orientation towards stimulation and risk was directly manifested in his or her responses to the MDSI items. The higher a sensation seeking tendency, the higher the tendency to adopt a risky and high-velocity driving style—two manifestations of the need for stimulation and sensation during driving—and the lower the tendency to adopt a patient driving style—a style that is the opposite to a sensation seeking orientation.

Finally, extraversion was significantly and inversely related to dissociative and anxious driving styles. That is the higher the extraversion, the lower the tendency to adopt a dissociative driving style or to feel anxiety during driving. This pattern of findings fits extraverted people's tendency to take life easily and dismiss life hardships and difficulties. This personality orientation seems to reduce worries during

driving and the tendency to experience cognitive gaps and dissociative states due to these worries.

Overall, the findings reveal consistent and coherent patterns of associations between global personality traits and the eight driving styles. Importantly, partial correlations controlling for age, sex, and amount of weekly driving (in km) revealed an identical pattern of associations to that presented in Table 3. On this basis, we can conclude that the various MDSI factors tap meaningful constructs that are somewhat derived from global personality orientations.

2.2.4. Driving styles and self-reported driving behaviors

In the next step, we examined the associations between the eight driving styles and three self-reported driving behaviors: (a) the amount of weekly driving (in km), (b) the number of car accidents in which a participant reported he or she had been involved in, and (c) the number of driving offenses a participant reported he or she had committed.

Pearson correlations revealed that the amount of weekly driving was significantly and inversely related to anxious driving style, $r(326) = -0.26$, $P < 0.01$. That is the stronger a participant's anxiety while driving, the less the amount of driving he or she undertook. No other significant associations were found. As expected, persons who tend to feel driving stress tend to avoid driving. This behavioral manifestation of anxious driving style seems to strengthen the construct validity of this MDSI factor.

Partial correlations (controlling for age and weekly driving) revealed that involvement in car accidents was significantly and positively associated with angry, risky, and high-velocity driving styles, $r(324) = 0.22$, $P < 0.01$; $r(324) = 0.35$, $P < 0.01$; $r(324) = 0.26$, $P < 0.01$, and inversely associated with careful driving style, $r(324) = -0.23$, $P < 0.01$. That is the higher a participant's tendency to adopt angry, risky, or high-velocity driving styles, the higher the number of accidents he or she reported being involved in. Accordingly, the higher a participant's tendency to adopt a careful driving style, the lower the number of accidents he or she reported being involved in.

In order to examine the contribution of the eight MDSI factors to car accidents involvement, beyond the variance explained by sociodemographic variables (sex, age) and personality traits (extraversion, desire for control, self-esteem, sensation seeking), we performed a discriminant analysis in which all these 14 variables were entered into the model to discriminate between participants who reported being involved at least in one car accident and participants who reported being involved in no car accident. The standardized canonical coefficients revealed that beyond the contribution of sociodemographic and personality variables, some MDSI factors still contributed to the discriminant function (coefficients higher than 0.35). Specifically, the dissociative (0.36), risky (0.49), and high-velocity (0.62) driving styles made a unique contribution to car accidents involvement. Interestingly, after controlling for the MDSI factors, only age (0.73)

and sensation seeking (0.41) made unique contributions to car accidents involvement.

Partial correlations (controlling for age and weekly driving) revealed that the reported number of driving offenses was significantly and positively associated with risky and high-velocity driving, $r(324) = 0.19$, $P < 0.01$; $r(324) = 0.22$, $P < 0.01$. No other significant associations were found. That is the higher a participant's tendency to adopt risky or high-velocity driving styles, the higher the number of driving offenses he or she reported they had committed.

A multiple regression examining whether the driving style scores significantly predicted the number of driving offenses revealed that the set of the eight driving style scores significantly predicted driving offenses, $F(8, 317) = 3.65$, $P < 0.01$, and explained 12% of the variance of this variable. In addition, another regression that entered the eight MDSI factors, sociodemographic variables (sex, age) and personality traits (extraversion, desire for control, self-esteem, sensation seeking) as the predictors revealed that high-velocity driving style still made a unique significant contribution ($B = 0.33$, $P < 0.01$) beyond the variance explained by sociodemographic and personality scores. This regression also revealed that self-esteem was the single sociodemographic and personality variable that made a unique contribution after controlling for MDSI scores ($B = 0.31$, $P < 0.01$).

These findings present evidence supporting the validity of the MDSI factors. First, the MDSI factors significantly predicted self-reports of involvement in car accidents and the amount of driving offenses. Second, those styles that were expected to reflect maladaptive ways of driving, such as risky and high-velocity driving, significantly contributed to the involvement in car accidents and to the commission of driving offenses. Third, these styles still contributed to car accidents involvement and driving offenses after controlling for sociodemographic and personality variables.

2.2.5. Conclusions

In Study 1 we constructed a reliable and valid self-report scale tapping driving styles. Findings revealed that eight internally coherent factors of driving style underlie the items of this scale, and that these factors were significantly associated with relevant personality traits and sociodemographic characteristics. More importantly, findings indicated that these eight factors significantly predicted self-reports of involvement in car accidents and commission of driving offenses.

3. Study 2

The aim of Study 2 was to further examine the construct validity of the MDSI factors by focusing on maladaptive driving styles and their associations with negative affectivity. If anxious, dissociative, high-velocity, and angry driving styles are valid manifestations of maladaptive ways of driving, significant correlations should be found with measures

of global emotional maladjustment, such as trait anxiety and neuroticism.

3.1. Method

3.1.1. Participants

One hundred and fifty Israeli university and college students who had driving license and drove on a regular basis (86 women and 64 men, ranging in age from 19 to 45 years, mean = 23.50, S.D. = 4.01), volunteered to participate in the study and to complete a battery of self-report scales.

3.1.2. Measures and procedure

All participants completed the 44-item version of the MDSI (described in Study 1). In the current sample, a confirmatory factor analysis revealed that the 8 main factors explained 57.2% of the variance and replicated the factor structure described in Study 1. The internal consistency (Cronbach's α coefficients) of each of the eight MDSI factors was acceptable (ranging from 0.72 to 0.86). On this basis, we computed eight driving style scores by averaging the items of each MDSI factor. Trait anxiety was assessed with the trait form of the State-Trait Anxiety scale (Spilberger et al., 1970). This scale consisted of 20 statements tapping the cognitive, affective, and behavioral manifestations of anxiety. Participants rated the extent to which they agreed with each statement on a 4-point scale, ranging from "totally disagree" (1) to "totally agree" (4). The Cronbach's α coefficient for the 20 items in the current sample was high (0.90), allowing us to compute a trait anxiety score by averaging the 20 items. Neuroticism was assessed by a short 12-item Hebrew version of the Neuroticism subscale of the Eysenck Personality Inventory (Eysenck and Eysenck, 1967). In this version, participants were asked to rate their agreement with each item on a 5-point scale, ranging from "not at all" (1) to "very much" (5). A total neuroticism score was computed by averaging the 12 items (Cronbach's α of 0.89).

3.2. Results and discussion

A canonical correlation between the set of the eight driving style scores and the set of the two scores of negative affectivity revealed a significant association, $F(16, 272) = 3.28$, $P < 0.01$ and explained 21% of the variance. Pearson correlations revealed the following significant associations: trait anxiety was significantly and positively associated with anxious and dissociative driving styles, $r(148) = 0.36$, $P < 0.01$; $r(148) = 0.28$, $P < 0.01$, and inversely associated with careful and patient driving styles, $r(148) = -0.29$, $P < 0.01$; $r(148) = -0.25$, $P < 0.01$. That is the higher the trait anxiety, the higher the tendency to adopt anxious or dissociative driving styles and the lower the tendency to adopt careful or patient driving styles. Neuroticism was also significantly and positively associated with anxious and dissociative driving styles, $r(148) = 0.34$, $P < 0.01$; $r(148) = 0.29$, $P < 0.01$, and inversely associated with careful driv-

ing style, $r(148) = -0.28$, $P < 0.01$. That is the higher the neuroticism scores, the higher the tendency to adopt anxious or dissociative driving styles and the lower the tendency to adopt a careful driving style.

Overall, this pattern of findings strengthens our confidence in the validity of the MDSI as measuring adaptive and maladaptive driving styles. Both trait anxiety and neuroticism are basic signs of psychological maladjustment. Therefore, they should be positively associated with maladjusted styles of driving and inversely associated with well-adjusted ways of driving. As can be seen, the current findings provide strong support for this hypothesis. Whereas trait anxiety and neuroticism were positively associated with anxious and dissociative driving styles—two maladaptive driving styles, they were inversely associated with careful driving—a well-adjusted driving style.

4. General discussion

The purpose of this research was to highlight the need for an integrative multidimensional measure of driving styles and to examine the usefulness and validity of such a measure. Taken together, the two studies provide strong evidence for the value of distinguishing among different domains of driving style as well as for the internal validity and usefulness of the MDSI for explaining variations in adaptive and maladaptive driving behaviors. The correlations between the eight MDSI factors and the assessed personality traits further attest to the importance of distinguishing among different driving styles. Specifically, risky, dissociative, and high-velocity styles were most closely associated with a cluster of maladaptive traits and a history of reckless driving, whereas careful and patient styles were associated with adaptive aspects of personality and driving behavior.

Interestingly, although we hypothesized a construct including four central domains of driving style, a factor analysis of the MDSI provided strong evidence for an eight factor-solution. These eight factors cover the four expected driving style domains, while making more fine distinctions within each of the domains. Specifically, the reckless and careless driving style was represented by the risky and high-velocity MDSI factors; the anxious driving style was represented by the anxious, dissociative and distress-reduction MDSI factors; the angry and hostile driving style was directly represented by the angry MDSI factor; and the patient and careful driving style was represented by two conceptually related MDSI factors—the careful and patient factors. These eight internally coherent MDSI factors are compatible to our theoretical conceptualization as well as to previous studies on driving style. They also highlight the complexity of driving style and the need for a dimension-specific attitude when dealing with this phenomenon.

Several findings supported the validity of the MDSI factors. First, these factors were significantly associated with

self-reports of involvement in car accidents and driving offenses. Second, those MDSI factors that were theoretically expected to reflect maladaptive ways of driving, such as angry, risky and high-velocity driving, were significantly associated with self-reports of more frequent car accident involvement and commission of driving offenses. Third, the MDSI factor that was theoretically expected to reflect an adaptive way of driving (careful style) significantly contributed to less involvement in car accidents.

Findings concerning the association between MDSI factors and sociodemographic variables also strengthened our confidence in the construct validity of the MDSI. Our findings were in accordance with the literature, which reveals that women tend to exhibit more driving stress than men (e.g. Simon and Corbet, 1996) and that maladaptive driving seems to diminish with age (e.g. Glendon et al., 1996; Maycock et al., 1991). These two tendencies were clearly identified by the MDSI factors. First, women's driving stress was manifested in their relatively high scores in the anxious and dissociative MDSI factors. Second, the tendency of older people to adopt more adaptive ways of driving was manifested in their relatively high scores in careful and patient MDSI factors as well as in their low scores in angry, anxious, dissociative, risky, and high-velocity MDSI factors.

The observed variations in MDSI factors were also in accordance with general personality characteristics. Self-esteem, which represents a highly adaptive and healthy personality trait (Rosenberg, 1979), was positively associated with adaptive driving styles, and inversely associated with maladaptive driving styles. Whereas need for control was positively associated with the angry and careful MDSI factors, sensation seeking was directly manifested in the endorsement of risky and high-velocity MDSI factors, and extraversion was inversely related to the dissociative and anxious driving MDSI factors. Both trait anxiety and neuroticism, which are basic signs of psychological maladjustment, were positively associated with maladaptive driving styles.

Some limitations of the current studies should be noted. First, the studies relied on self-reports of a person's own driving behavior. Future studies should attempt to replicate the present findings using behavioral measures, such as observations of real-life driving or car-simulator driving, and adopting a multi-method measurement approach. Second, no systematic attempt was made to examine the validity and usefulness of the MDSI in specific at risk populations. Further examinations should focus on specific at risk groups, such as recidivist traffic offenders, young drivers, etc. Despite these limitations, the current research provides important evidence regarding the usefulness of the MDSI for explaining reckless driving behavior. Future studies should assess the associations of the MDSI factors with other relevant individual-differences factors, such as locus of control and hardiness, and examine the dynamics of driving style in different situational contexts, (e.g. the presence of peers or adults in the car).

In conclusion, it has been claimed that some 90% of road-traffic accidents are caused by driver error (Lewin, 1982). The real challenge is therefore to provide a better understanding of the role of human factors in the causation of road accidents and consequently to develop effective countermeasures. These countermeasures may take the form of improved driver training and testing, education campaigns aimed at changing driving practices, legislation to control driver behavior, and improvements in the design of road systems and vehicles (Elander et al., 1993). We believe that the MDSI scores can be taken as driving-specific factors within a comprehensive model of reckless driving, as well as working guidelines for the construction of effective countermeasures.

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